AFRICA FINTECH FOUNDRY

ACCESS YOUTH TECH FORGE

Sandbox Participant Platform Guide

Powered by access

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This guide covers everything a team needs to run its hackathon workload on the Forge 2026 sandbox: access, deployment, the platform APIs, limits and support. Read sections 1 to 4 before your first deploy; use sections 5 to 8 as reference.

> Conventions: `teamXX` is your team ID (for example `team07`). Commands assume macOS or Linux; Windows users run them in WSL or Git Bash. Values in angle brackets, such as `<team-key>`, are yours to fill in.

1 Quick start

From zero to a live URL. Prerequisites: `kubectl` 1.30 or later, Docker, and the `teamXX.yaml` kubeconfig your team lead received.

TERMINAL

```
# 1. Point kubectl at your namespace
export KUBECONFIG=~/forge/teamXX.yaml
kubectl get pods                    # expect: No resources found in teamXX namespace.

# 2. Build for the cluster's architecture and push
docker build --platform linux/amd64 -t <registry>/<image>:v1 .
docker push <registry>/<image>:v1

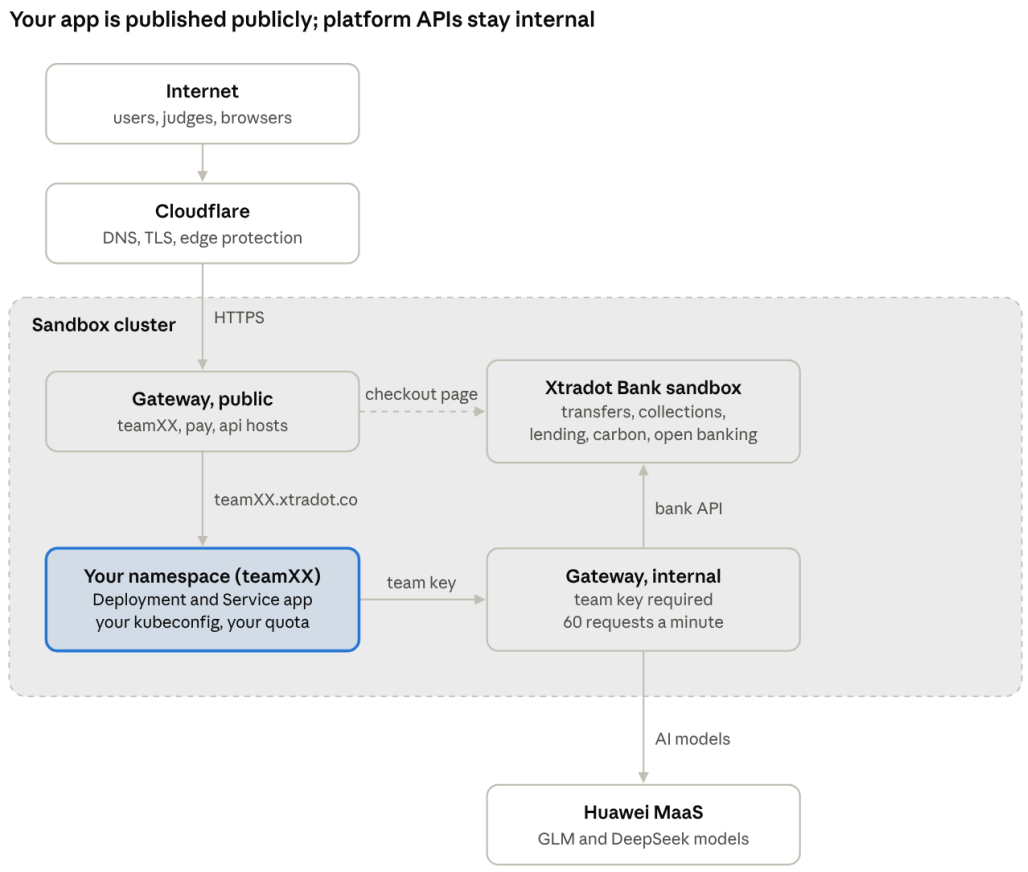
# 3. Deploy (manifest in section 4.2)
kubectl apply -f app.yaml
kubectl rollout status deployment/app

# 4. Verify
curl -I https://teamXX.xtradot.co
```

If step 4 returns `200`, you're live. Platform API access (bank sandbox, Huawei MaaS) is covered in sections 5 and 6.

2 Environment overview

Each team gets a dedicated Kubernetes namespace on a managed cluster. Inbound traffic arrives through Cloudflare and the public gateway listener, which routes `teamXX.xtradot.co` to the `app` service in your namespace. The platform APIs sit behind a second, internal listener that accepts traffic only from team namespaces and requires your team key.



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ENDPOINT	REACHABLE FROM	PURPOSE
`https://teamXX.xtradot.co`	Internet	Your published app (`app` Service, port 80)
`http://bank-api.ingress-apisix.svc.cluster.local/v1`	Team namespaces	Xtradot Bank sandbox API (section 5)
`http://huawei-maas.ingress-apisix.svc.cluster.local/openai/v1`	Team namespaces	Huawei MaaS, OpenAI-compatible (section 6)
`https://pay.xtradot.co/pay/<token>`	Internet	Hosted checkout page returned by the bank API
`http://app.teamXX.svc.cluster.local`	Inside the cluster	Your app's internal address, for webhooks

Platform facts: nodes are `linux/amd64`; TLS terminates at Cloudflare and the gateway, so your app serves plain HTTP; `LoadBalancer` and `NodePort` Services are blocked by quota; namespaces are isolated from each other at the network layer.

3 Access and credentials

Each team lead receives one file through a private channel: the team's kubeconfig. The team key is already provisioned in your namespace, so it never travels by message.

CREDENTIAL	GRANTS	HOW YOU GET IT	SCOPE
Kubeconfig `teamXX.yaml`	Deploy, update, read logs and exec into pods	Sent privately to the team lead	Your namespace only; other namespaces return `Forbidden`
Team key	Calls to the bank sandbox and Huawei MaaS	Pre-provisioned as secret `huawei-gateway` in your namespace	Inside the cluster only; useless from the internet

3.1 Kubeconfig

```
TERMINAL
mkdir -p ~/forge && mv ~/Downloads/teamXX.yaml ~/forge/ && chmod 600 ~/forge/teamXX.yaml
export KUBECONFIG=~/forge/teamXX.yaml
kubectl auth can-i create deployments   # yes
kubectl auth can-i get pods -n team01   # no (unless you are team01)
```

To share access within your team, give each member the same file through a password manager. Don't commit it to Git.

3.2 Team key

The key lives in the secret `huawei-gateway` (field `GATEWAY_KEY`) in your namespace. The reference manifest in 4.2 injects it into your container as the environment variable `GATEWAY_KEY`; no further setup is needed. It's the same key for both APIs; only the header differs: `x-api-key` for the bank sandbox, `apikey` for Huawei MaaS.

To read it, for example for manual testing (section 8.1):

```
TERMINAL
kubectl get secret huawei-gateway -o jsonpath='{.data.GATEWAY_KEY}' | base64 --decode; echo
```

Your kubeconfig can only read secrets in your own namespace; requests for another team's secret return `Forbidden`. Every call made with your key is attributed to your team in the gateway logs.

Never bake the key into an image, source code or a public repository. If it leaks, report it (section 9); the platform team rotates it in place, and pods pick up the new value on their next restart (`kubectl rollout restart deployment/app`).

3.3 Compromise or loss

Report a leaked kubeconfig or key immediately. The platform team revokes the credential and issues a replacement. Workloads keep running during rotation; only `kubectl` access and API calls with the old key stop.

4 Deploying workloads

4.1 Image requirements

- Built for `linux/amd64`. ARM images built on Apple Silicon without `--platform` fail with `exec format error`.
- Listens on `0.0.0.0`, not `127.0.0.1`, on a fixed port.
- Exposes a cheap health endpoint (for example `GET /healthz` returning `200`).
- Pulled from a registry the cluster can reach: Docker Hub, GHCR or similar. For a private image, create a pull secret once and reference it as `imagePullSecrets`:

```
TERMINAL
kubectl create secret docker-registry regcred \
  --docker-server=ghcr.io --docker-username=<user> --docker-password=<token>
```

4.2 Reference manifest

Adjust `image` and `containerPort` (`8080` here), then `kubectl apply -f app.yaml`.

```
YAML
apiVersion: apps/v1
kind: Deployment
metadata:
  name: app
spec:
  replicas: 1
  selector:
```

```
    matchLabels: { app: app }
  template:
    metadata:
      labels: { app: app }
    spec:
      # imagePullSecrets: [{ name: regcred }]
      containers:
      - name: app
        image: <registry>/<image>:v1
        ports:
        - containerPort: 8080
        envFrom:
        - secretRef: { name: huawei-gateway, optional: true }
        readinessProbe:
          httpGet: { path: /healthz, port: 8080 }
          periodSeconds: 5
        resources:
          requests: { cpu: 50m, memory: 128Mi }
          limits: { cpu: 500m, memory: 512Mi }
---
apiVersion: v1
kind: Service
metadata:
  name: app
spec:
  selector: { app: app }
  ports:
  - port: 80
    targetPort: 8080
```

4.3 Publishing contract

The gateway publishes exactly one Service per team. No ticket or DNS change is needed: the route exists already.

REQUIREMENT	VALUE
Service name	`app`
Service port	`80` (any `targetPort`)
Public URL	`https://teamXX.xtradot.co`
Protocol to your pod	Plain HTTP; TLS is terminated upstream

Other Services (a database, a worker, a second backend) stay internal and are reachable from your own pods by name, for example `http://worker`. If you need a second public host, ask the platform team.

4.4 Day-two operations

TERMINAL

```
kubectl get pods                              # status
kubectl logs deploy/app -f                    # follow logs
kubectl logs deploy/app --previous            # logs from the last crash
kubectl exec -it deploy/app -- sh             # shell, if the image has one

kubectl set image deployment/app app=<registry>/<image>:v2
kubectl rollout status deployment/app
kubectl rollout undo deployment/app           # roll back
```

Tag every release (`v1` , `v2` , or a Git SHA). Reusing `latest` makes rollbacks unpredictable and may not pull the new image at all.

5 Xtradot Bank sandbox API

A simulated Nigerian bank. Flows mirror production behaviour (name enquiry before transfer, NIP session IDs, fees and VAT, KYC limits, signed webhooks), but all data is synthetic and nothing connects to NIBSS, NIMC or a real core banking system. Each team has an isolated

copy of the test data. Using this API is optional. The authoritative contract is the *Sandbox API Mock Specification v2*; this section is the operational summary.

5.1 Conventions

ITEM	VALUE
Base URL	`http://bank-api.ingress-apisix.svc.cluster.local/v1` (team namespaces only)
Auth	`x-api-key: <team-key>` on every request
Idempotency	`Idempotency-Key: <uuid>` required on every POST that moves money or creates a record; same key and body replays the original response with `Idempotent-Replayed: true`; same key, different body returns `409`
Amounts	NGN, two decimals
Timestamps	ISO 8601, UTC (`Z`)
Pagination	`?page=1&per_page=20` , max 100; lists return `meta`
Rate limit	60 requests a minute per team, shared with Huawei MaaS; `429` beyond
Forced outcomes	`x-test-outcome: successful | failed | pending | timeout | reversed`

Responses use one envelope:

JSON

```
{"status":"success","message":"...","data":{},"meta":null,"request_id":"req_7f3a9c21"}
{"status":"error","error":{"code":"INVALID_REQUEST","message":"...","field":"amount"},"request_id":"req_7f3a9c22"}
```

Quote `request_id` in any support request.

5.2 Transfers

Every transfer is two calls: name enquiry, then the transfer carrying the returned `name_enquiry_ref`. Display the resolved name to the user before sending.

TERMINAL

```
BASE=http://bank-api.ingress-apisix.svc.cluster.local/v1

curl -s -X POST $BASE/transfers/interbank/name-enquiry \
  -H "x-api-key: $GATEWAY_KEY" -H 'Content-Type: application/json' \
  -d '{"destination_bank_code":"000013","account_number":"0123987001"}'

curl -s -X POST $BASE/transfers/interbank \
  -H "x-api-key: $GATEWAY_KEY" -H 'Content-Type: application/json' \
  -H "Idempotency-Key: $(uuidgen)" \
  -d '{"source_account":"0123456790","destination_bank_code":"000013",
       "destination_account":"0123987001","name_enquiry_ref":"<ref>",
       "amount":75000.00,"narration":"Generator repairs","reference":"TRF-0002"}'
```

Intrabank (Xtradot Bank to Xtradot Bank) uses `/transfers/intra/name-enquiry` and `/transfers/intra`, completes in the call (`201`) and carries no transfer fee. Interbank returns `202 pending` and settles in about 5 seconds.

Enforced rules: a `name_enquiry_ref` is single-use and expires after 10 minutes; destination must match the enquiry; intra and NIP refs aren't interchangeable; the source is debited amount plus charges; dormant and frozen beneficiaries fail at enquiry; PND sources can't send; KYC single and daily limits apply (tier 1: ₦50,000 / ₦300,000; tier 2: ₦100,000 / ₦500,000; tier 3: ₦5,000,000 / ₦25,000,000; corporate: ₦10,000,000 / ₦100,000,000).

> **On `504 TIMEOUT` , never resend.** Query `GET /transfers/{reference}` or `POST /transfers/interbank/requery` until the status is final. Resending is the classic double-debit defect, and it is assessed.

Charges. Every transfer response returns `fee`, `vat`, `stamp_duty` and `total_debit`; `GET /tariff` returns the active schedule. The event starts on the sandbox schedule (NIP ₦10 / ₦25 / ₦50 by band, plus 7.5% VAT). The platform team may switch to the CBN 2026 schedule for demo day (NIP ₦0 up to ₦5,000, ₦10 up to ₦50,000, ₦50 above, plus VAT, and ₦50 stamp duty on any transfer of ₦10,000 or more). Read charges from the response; don't hard-code them.

5.3 Endpoint catalogue

DOMAIN	ENDPOINTS
Reference data	GET /banks GET /banks/{nip_code}/availability GET /tariff
Transfers	POST /transfers/intra/name-enquiry POST /transfers/intra POST /transfers/interbank/name-enquiry POST /transfers/interbank GET /transfers/{reference} POST /transfers/interbank/requery POST /transfers/bulk GET /transfers/bulk/{batch_reference}
Accounts	GET /accounts/{account_number}/balance GET /accounts/resolve GET /transactions GET /transactions/{reference}
Wallets	POST /wallets GET /wallets/{wallet_id} POST /wallets/{wallet_id}/fund POST /wallets/transfer
Collections	POST /payments/checkout POST /payments/{reference}/card POST /payments/{reference}/otp POST /payments/{reference}/simulate-transfer + /simulate-ussd + /simulate-qr GET /payments/{reference}/verify POST /pos/transactions
Settlement	GET /settlements POST /reconciliation
Virtual accounts	POST /virtual-accounts GET /virtual-accounts/{account_number} POST /virtual-accounts/{account_number}/simulate-inflow
Identity	POST /identity/bvn/verify POST /identity/nin/verify
Open banking	POST /open-banking/consents POST /open-banking/consents/{id}/authorise GET DELETE /open-banking/consents/{id} GET /open-banking/accounts/{account_number} + /balance + /statement (all reads take ?consent_id=)
Lending	POST /loans/applications GET /loans/{loan_id} POST /loans/{loan_id}/accept + /disburse + /repayments

DOMAIN	ENDPOINTS
Carbon registry	POST /carbon/projects GET /carbon/projects/{id} POST /carbon/projects/{id}/mrv GET /carbon/mrv/{id} POST /carbon/mrv/{id}/review POST /carbon/credits/retire GET /carbon/certificates/{id}
Webhooks	POST /webhooks GET /webhooks DELETE /webhooks/{id} POST /webhooks/test GET /webhooks/deliveries POST /webhooks/deliveries/{event_id}/replay
Team admin	POST /admin/reset POST /admin/accounts/{n}/fund POST /admin/accounts/{n}/status POST /admin/banks/{nip_code}/status POST /admin/settlements/run POST /admin/loans/{id}/advance-to-due

Not implemented: merchant onboarding, KYC tiering and the specification's optional services (mandates, split and escrow, refunds and disputes, cards, cross-border, bills, risk). These return `404 NOT_FOUND`.

Collections notes: `payment_url` opens a public hosted checkout at `pay.xtradot.co`, which redirects to your `callback_url` with `reference` and `status`. Always confirm with `GET /payments/{reference}/verify` before giving value. Merchant fees: card and POS 1.5% capped at ₦2,000; transfer, USSD and QR ₦50 flat.

5.4 Test data

Team-owned accounts, Xtradot Bank (NIP `000014`, CBN `044`):

ACCOUNT	HOLDER	TYPE	TIER	OPENING BALANCE (₦)	BVN
0123456789	Adaeze Chiamaka Okonkwo	Savings	3	1,500,000.00	22200000001
0123456790	Babatunde Olumide Adeyemi	Current	3	5,000,000.00	22200000002
0123456791	Musa Ibrahim Abdullahi	Savings	2	250,000.00	22200000003
0123456792	Efosa Godwin Osagie	Savings	1	40,000.00	22200000004
0123456793	Mama Nkechi Provisions Ltd	Corporate	n/a	12,000,000.00	n/a

Beneficiaries:

ACCOUNT	HOLDER	BANK (NIP)	STATE
0987654321	Ngozi Adaobi Eze	Xtradot Bank (000014)	active
0987654322	Aisha Hauwa Bello	Xtradot Bank (000014)	active
0987654399	Obinna Kelechi Okafor	Xtradot Bank (000014)	dormant
0123987001	Chukwuemeka Ifeanyi Nwosu	GTBank (000013)	active
0123987099	Garba Sani Danjuma	GTBank (000013)	PND
0123987002	Funmilayo Abiola Ogunleye	Zenith (000015)	active
0123987003	Zainab Halima Yusuf	First Bank (000016)	active
0123987004	Tamunotonye Briggs	UBA (000004)	active

ACCOUNT	HOLDER	BANK (NIP)	STATE
0123987098	Blessing Oghenekaro Edeh	Wema (000017)	frozen
8031234506	Ifeoluwa Grace Adebayo	OPay (100004)	active
8051234507	Iya Bisi Foods Enterprises	Moniepoint (090405)	active
0123987005	Arewa Solar Co-operative Society	Zenith (000015)	carbon payouts
0123987006	Ogoja Community Forest Trust	First Bank (000016)	carbon payouts

Other fixtures: BVNs `22200000001` to `22200000004` and NINs `70000000001` to `70000000004` match the four individuals above (DOBs 1998-04-12, 1985-09-30, 1992-01-17, 2001-06-05). Merchants: `mer_204` Mama Nkechi Provisions, `mer_205` Kano Agro Hub, `mer_206` Ọ̀ṣun Threads, `mer_900` Forge Carbon Registry. Wallets: `wal_adaeze_01` (₦20,000), `wal_musa_01` (₦5,000). Every team account carries seeded October 2026 history for statements and credit scoring. OTP for cards and consents: `123456`.

Test cards (any future expiry, CVV `123`, PIN `1234`): `5399830000000008` succeeds; `5399830000000016` declines (insufficient funds); `5399830000000024` fails OTP.

5.5 Trigger values

TRIGGER	APPLIES TO	RESULT
Account `0000000001`	Name enquiry, bulk	`ACCOUNT_NOT_FOUND`
Account `0000000002`	Name enquiry	`TIMEOUT` after 10 s
Bank code `999999`	Interbank enquiry	`BANK_UNAVAILABLE`
Amount ending `.01`	Transfers, card, POS	`INSUFFICIENT_FUNDS`
Amount ending `.02`	Transfers	`TIMEOUT`, then failed and reversed after 30 s
Amount ending `.03`	Transfers	Pending 2 min, then successful
Amount ending `.04`	Interbank	Successful, then reversed after about 60 s
Amount ending `.05`	Transfers	First webhook delivery withheld
Reference prefix `DUP-`	Transfers, payments	`DUPLICATE_REQUEST`
BVN `22299999999`	Identity, loans	No match / declined
NIN `70099999999`	Identity	No match

Amounts ending `.00` never trigger.

5.6 Webhooks

Register an HTTPS public URL or your internal address (`http://app.teamXX.svc.cluster.local/...`). `webhook_secret` is returned once at registration.

TERMINAL

```
curl -s -X POST $BASE/webhooks -H "x-api-key: $GATEWAY_KEY" -H 'Content-Type: application/json' \
  -d '{"url":"http://app.teamXX.svc.cluster.local/hooks/forge","events":["*"]}'
```

Delivery headers: `x-forge-event`, `x-forge-timestamp`, `x-forge-signature: sha256=<hex HMAC-SHA256 of "{timestamp}.{raw_body}">`. Respond `2xx` within 10 s; failures retry at 1 min, 5 min, 15 min, 1 h and 6 h. De-duplicate on `event_id`; reject timestamps older than 5 minutes.

```python
import hashlib, hmac, time

def verify(raw: bytes, ts: str, sig: str, secret: str) -> bool:
    if abs(time.time() - int(ts)) > 300:
        return False
    mac = hmac.new(secret.encode(), ts.encode() + b"." + raw, hashlib.sha256).hexdigest()
    return hmac.compare_digest("sha256=" + mac, sig)
```

Events: `transfer.pending|successful|failed|reversed`, `transfer.bulk.completed`, `payment.pending|successful|failed|partial|overpaid`, `virtual_account.credited`, `settlement.completed`, `consent.authorised|revoked`, `loan.disbursed|repayment_received|overdue|closed`, `carbon.mrv.verified|rejected`, `carbon.credits.retired`, `webhook.test`.

5.7 Error codes

CODE	HTTP	MEANING
`INVALID_REQUEST`	400	Malformed or missing field; `field` names it
`UNAUTHORIZED`	401	Missing or invalid key (the gateway may return its own short 401 body)
`INSUFFICIENT_FUNDS`	402	Balance below amount plus charges
`ACCOUNT_RESTRICTED`	403	Source is PND, dormant or frozen
`CONSENT_REQUIRED`	403	No active open banking consent for the scope
`ACCOUNT_NOT_FOUND` `NOT_FOUND`	404	Unknown account, record or endpoint
`DUPLICATE_REQUEST`	409	Reused key, reference or enquiry ref
`NAME_ENQUIRY_EXPIRED` `NAME_ENQUIRY_MISMATCH`	409	Enquiry older than 10 min, or destination differs
`ACCOUNT_DORMANT` `BENEFICIARY_FROZEN`	422	Beneficiary can't receive
`LIMIT_EXCEEDED` `BVN_MISMATCH` `INVALID_OTP`	422	KYC limit, identity mismatch, wrong OTP
`RATE_LIMITED`	429	Over 60 requests a minute
`BANK_UNAVAILABLE`	503	Beneficiary bank down
`TIMEOUT`	504	Outcome unknown: requery, never resend

`POST /admin/reset` restores your team's opening state at any time; it doesn't affect other teams.

6 Huawei MaaS

Huawei Cloud model inference, OpenAI-compatible. Huawei has issued one sandbox key per team; the gateway holds them and injects yours, so you authenticate with your team key only. Where Huawei's own participant guide asks for an API key, use the values below instead.

SETTING	VALUE
Base URL	`http://huawei-maas.ingress-apisix.svc.cluster.local/openai/v1`
API key (SDK field)	Any placeholder, for example `unused`

SETTING	VALUE
Auth header	`apikey: <team-key>`
Models	`glm-5.3` , `glm-5.2` , `glm-5.1` , `deepseek-v4.1-flash-1017` , `deepseek-v4.1-flash` , `deepseek-v4-flash`

Model availability depends on what Huawei has enabled; if a model returns not found, use another from the list.

PYTHON

```python
import os
from openai import OpenAI

client = OpenAI(
    base_url="http://huawei-maas.ingress-apisix.svc.cluster.local/openai/v1",
    api_key="unused",
    default_headers={"apikey": os.environ["GATEWAY_KEY"]},
)

resp = client.chat.completions.create(
    model="glm-5.2",
    messages=[{"role": "user", "content": "Classify this transaction narration: 'POS SALE - KANO AGRO HUB'"}],
    extra_body={"chat_template_kwargs": {"thinking": False}},
)
print(resp.choices[0].message.content)
```

JAVASCRIPT

```javascript
import OpenAI from "openai";

const client = new OpenAI({
  baseURL: "http://huawei-maas.ingress-apisix.svc.cluster.local/openai/v1",
  apiKey: "unused",
  defaultHeaders: { apikey: process.env.GATEWAY_KEY },
});
```

Usage guidance. Sandbox keys may carry quotas, so treat model calls as a budget:

- Disable reasoning (`chat_template_kwargs: {thinking: false}`) unless you need it. GLM models reason by default and `max_tokens` doesn't cap it; in platform testing, a one-line reply used 137 tokens with reasoning and 18 without.
- Don't call models on page load, in loops, or from unauthenticated public endpoints without your own throttling.
- Cache responses you reuse.
- Requests can take several seconds; the gateway allows up to 120 s. Set client timeouts accordingly and stream where your UX benefits.

7 Quotas, limits and operating policies

7.1 Namespace quota

RESOURCE	LIMIT
Pods	10
CPU requests / limits (namespace total)	400m / 2 cores
Memory requests / limits (namespace total)	768Mi / 2Gi
Services	5 (`LoadBalancer` and `NodePort` : 0)
Persistent volumes	1, up to 10Gi
Container defaults if unset	requests 50m / 64Mi, limits 500m / 512Mi

Check consumption with `kubectl describe resourcequota team-quota` . A pod rejected with `exceeded quota` needs smaller `resources.requests` , or fewer pods.

7.2 Platform API limits

LIMIT	VALUE
Requests per team	60 a minute, bank sandbox and Huawei MaaS combined
Gateway timeout, bank sandbox	30 s
Gateway timeout, Huawei MaaS	120 s

7.3 Policies

- **Synthetic data only.** No real names, account numbers, BVNs, NINs, card numbers or other personal data anywhere in the sandbox, including logs and seed files.
- **Public by default.** Anything served at `teamXX.xtradot.co` is reachable by anyone with the link. Gate what shouldn't be public.
- **Isolation.** Don't probe other namespaces or platform components. Network policy blocks traffic between team namespaces.
- **Observability.** Gateway requests are logged and metered per team for capacity planning, support and judging.
- **Sandbox state is volatile.** Bank sandbox data lives in memory. A platform restart restores every team to the opening state; planned restarts are announced in advance. Don't rely on sandbox data surviving overnight, and keep your own records in your app.
- **End of event.** Kubeconfigs and team keys are revoked when the event closes; export anything you need before then.

8 Troubleshooting

8.1 Testing the platform APIs

The bank sandbox and Huawei MaaS don't answer requests from your workstation; calls time out by design. Run test requests from a throwaway pod in your namespace:

TERMINAL

```
kubectl run curl --image=curlimages/curl --restart=Never -- sleep 3600
kubectl wait pod/curl --for=condition=Ready --timeout=60s

KEY=$(kubectl get secret huawei-gateway -o jsonpath='{.data.GATEWAY_KEY}' | base64 --decode)
kubectl exec curl -- curl -s -H "x-api-key: $KEY" \
  http://bank-api.ingress-apisix.svc.cluster.local/v1/accounts/0123456789/balance

kubectl delete pod curl
```

8.2 Symptom reference

SYMPTOM	LIKELY CAUSE	ACTION
`Forbidden` from kubectl	Command targets another namespace or a resource your role doesn't manage	Drop `-n`; check `kubectl auth can-i <verb> <resource>`
`ImagePullBackOff`	Wrong image reference, or private image without `regcred`	`kubectl describe pod <pod>`; see 4.1
`CrashLoopBackOff`	Application exits on start	`kubectl logs deploy/app --previous`
`exec format error`	ARM image	Rebuild with `--platform linux/amd64`
Pod `Pending`, `exceeded quota`	Requests above namespace quota	Lower requests; see 7.1
`502` / `503` on `teamXX.xtradot.co`	No `app` Service, wrong `targetPort`, or pod not ready	`kubectl get svc,endpoints app`; check readiness probe
`404 Route Not Found` on your URL	Hostname typo	Use `teamXX.xtradot.co` exactly
`401` from a platform API	Missing key, or wrong header	`x-api-key` for bank, `apikey` for Huawei
`429`	Rate limit	Back off, batch or cache

SYMPTOM	LIKELY CAUSE	ACTION
Platform API times out	Called from outside the cluster, or from a pod outside your namespace	Call from your namespace; see 8.1
`404 NOT_FOUND` with *not available in the sandbox build*	Optional service not implemented	See 5.3
Bank data back at opening balances	`/admin/reset` was called, or a platform restart	Re-seed your flow; see 7.3

Before escalating, collect: team ID, the exact command or request, the full error, the `request_id` if the bank API returned one, and the output of `kubectl get pods` and `kubectl describe pod <pod>`.

9 Support and change log

9.1 Support

CHANNEL	USE FOR
[Support channel, TBC]	Access problems, deployment issues, platform API errors
[Escalation contact, TBC]	Leaked credentials, platform-wide outages

Include the details listed at the end of section 8. Never paste a kubeconfig or team key into a support message; refer to it by team ID.

Out of scope: debugging your application code, choosing your stack, and judging decisions. Mentors cover the first two; the organizers cover the third.

9.2 Change log

VERSION	DATE	CHANGE
1.0	9 Nov 2026	Initial release for the build phase